# lec522: Final Review 2

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## 1 Ex. 3 PCA

$$Z = aX_1 + bX_2$$

- Given four PCA basis vector choices (assume the first PCA vector), which one is actually from the data?
- Given a choice $PC_1$, project $X_1, \ldots, X_n$ onto the $PC$, $Z_1 \ldots Z_n$. Check the variation of $Z_1 \ldots Z_n$.
- PCs are variation maximizing, a correct $PC_1$ is a basis whose projections are spread out.
- In our example, the $PC$ follows

$$PC_1 = \begin{bmatrix} \text{Age} \\ \text{AnnualIncome} \\ \text{SpendingScore} \\ \text{OnlineVisits} \end{bmatrix}$$

- In our example, `Age` and `AnnualIncome` have a positive correlation, not a negative correlation. So the PC

$$PC_1 = \begin{bmatrix} 0.56 \\ -0.78 \\ 0.32 \\ 0.03 \end{bmatrix}$$

can be eliminated. That leaves the correct answer:

$$PC_1 = \begin{bmatrix} 0.56 \\ 0.61 \\ -0.31 \\ 0.02 \end{bmatrix}$$

- Furthermore, we know that `OnlineVisits` has the least variance, so it should have the smallest contribution to the principal component. So the PC

$$PC_1 = \begin{bmatrix} 0.56 \\ -0.78 \\ 0.32 \\ 0.03 \end{bmatrix}$$

  can also be eliminated.

## 2 Ex. 4 Building a Model

- ID should not be included inside the model - IDs are independent and spurious trends can be found that are not easily understandable
- Standardize outliers, or can convert to percentile
- One-hot encoding creates one column per category, label encoding assigns each label a number.
- Label encoding - model might assume that there is an ordering, when there isn't
- One hot encoding - creates a column per unique level of the categorical variable, can result in too many columns to fit to (sparse matrix)